GATE ME 90 Days Challenge

Day 87/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Full Syllabus	Full-Length Mock 7	10 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on a mixed full-syllabus warm-up for the seventh full-length mock: antonyms, consecutive odd numbers, LU decomposition, the uniform distribution, angular impulse, cantilever deflection under a UDL, elongation of a bar under its own weight, the pantograph, flange-coupling bolts, the critical velocity for laminar flow, fin effectiveness, the mean temperature of heat addition, the heat-affected zone in welding, blanking work and multifactor productivity. Try them in **60 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 11 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 65 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

Select the word that is most nearly OPPOSITE in meaning to OBDURATE.

(A) Stubborn
(B) Unrepentant
(C) Yielding
(D) Inflexible

Q2 NAT

The average of five consecutive odd numbers is 27. The largest of these numbers is _____

Q3 MCQ

The matrix A with rows (2, 1) and (4, 5) is written as A = LU, where L is lower triangular with 1s on its diagonal (Doolittle's method). The upper triangular matrix U has rows

(A) (2, 1) and (0, 5)
(B) (2, 1) and (0, 3)
(C) (1, 0) and (2, 1)
(D) (2, 1) and (0, −3)

Q4 NAT

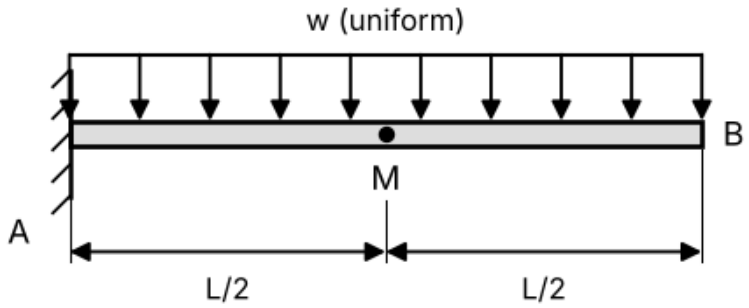
A random variable X is uniformly distributed over the interval [2, 8]. The variance of X is _____

Q5 NAT

A flywheel with a moment of inertia of 2 kg·m² rotates at 300 rpm. A constant braking torque brings it to rest in 10 s. The braking torque (in N·m, round off to two decimal places) is _____

Q6 MCQ

A cantilever AB of length L (fixed at A, free at B), shown in the figure, carries a uniformly distributed load over its whole length. The ratio of the deflection at its mid-span M to the deflection at its free end B is



(A) 5/16
(B) 1/2
(C) 11/48
(D) 17/48

Q7 NAT

A uniform steel rod 100 m long hangs vertically from its upper end and carries no load other than its own weight (density 7850 kg/m³, E = 200 GPa, g = 9.81 m/s²). The elongation of the rod (in mm, round off to two decimal places) is _____

Q8 MCQ

A pantograph is a linkage that

(A) copies a path to an enlarged or reduced scale, its links forming a parallelogram
(B) gives a quick return stroke to the ram of a shaper
(C) converts uniform rotation into simple harmonic motion
(D) transmits rotation between two parallel offset shafts

Q9 MCQ

A flange coupling transmits a torque of 3 kN·m through 6 bolts arranged on a pitch circle of 150 mm diameter. If the allowable shear stress in the bolts is 40 MPa, the bolt diameter needed is about

(A) 10.3 mm
(B) 20.6 mm
(C) 14.6 mm
(D) 7.3 mm

Q10 NAT

Oil of kinematic viscosity 1.0×10^{-4} m²/s flows through a pipe of 50 mm diameter. Taking the critical Reynolds number as 2000, the largest mean velocity at which the flow is certain to remain laminar (in m/s) is _____

Q11 MCQ

A very long aluminium pin fin (k = 200 W/m·K) of 5 mm diameter is exposed to air with h = 25 W/m²·K. Its effectiveness (heat transfer with the fin ÷ heat transfer from the bare base area) is about

(A) 40
(B) 80
(C) 160
(D) 8

Q12 NAT

Water is heated at constant pressure from 300 K to 400 K (constant specific heat). The thermodynamic mean temperature of heat addition, defined so that $Q = T_m \times \Delta S$, is (in K, round off to one decimal place) _____

Q13 MCQ

When a medium-carbon (hardenable) steel is arc welded without preheating, the heat-affected zone next to the fusion line often becomes hard and brittle and may crack. The main reason is that

(A) the filler metal always has a higher carbon content than the plate

(B) the arc adds nitrogen to the plate, forming hard nitrides

(C) the zone recrystallises into very coarse ferrite grains

(D) the zone is austenitised and then cooled fast, forming martensite

Q14 NAT

A 40 mm diameter blank is cut from a 3 mm thick sheet whose shear strength is 300 MPa. The punch penetrates 40 % of the sheet thickness before the blank breaks away. Taking the punch force as constant at its maximum value over the penetration, the work done in blanking (in J, round off to one decimal place) is _____

Q15 MCQ

A plant produces 1200 units in a week using 400 labour-hours paid at ₹200 per hour and materials costing ₹60 000. Its multifactor productivity (units per ₹1000 of labour and material input) is about

(A) 8.57

(B) 15.0

(C) 20.0

(D) 3.00

Answer Key

Q	Type	Answer
Q1	MCQ	(C) Yielding
Q2	NAT	31
Q3	MCQ	(B) (2, 1) and (0, 3)
Q4	NAT	3
Q5	NAT	6.28 (accept 6.22 to 6.34)
Q6	MCQ	(D) 17/48
Q7	NAT	1.93 (accept 1.91 to 1.94)
Q8	MCQ	(A) copies a path to an enlarged or reduced scale, its links forming a parallelogram
Q9	MCQ	(C) 14.6 mm
Q10	NAT	4
Q11	MCQ	(B) 80
Q12	NAT	347.6 (accept 346.5 to 348.7)
Q13	MCQ	(D) the zone is austenitised and then cooled fast, forming martensite
Q14	NAT	135.7 (accept 134.5 to 136.9)
Q15	MCQ	(A) 8.57

Detailed Solutions

Q1 **Answer: (C) Yielding**

Concept: 'Obdurate' describes a person who stubbornly refuses to change an opinion or course of action.

Stubborn, unrepentant and inflexible are all close to it in meaning.

The opposite is **yielding**, option (C).

Exam tip: In antonym questions three options are often synonyms of the given word; eliminate them first.

Q2 **Answer: 31**

Concept: For numbers in arithmetic progression, the average equals the middle term.

The middle (third) number is 27, and consecutive odd numbers differ by 2.

The numbers are 23, 25, 27, 29, 31, so the largest is **31**.

Exam tip: For an odd count of equally spaced numbers, start from the middle term instead of setting up an equation.

Q3 **Answer: (B) (2, 1) and (0, 3)**

Concept: Doolittle's LU decomposition is Gaussian elimination: U is the reduced matrix, and L stores the multipliers.

The multiplier for row 2 is $l_{21} = 4/2 = 2$. Row 2 $-$ 2 $\times$ row 1 = (4 $-$ 4, 5 $-$ 2) = (0, 3).

So U has rows **(2, 1) and (0, 3)**, option (B), and L has rows (1, 0) and (2, 1).

Exam tip: Check: det A = 2 $\times$ 5 $-$ 1 $\times$ 4 = 6 = product of the diagonal of U (2 $\times$ 3).

Q4 **Answer: 3**

Concept: For a uniform distribution on [a, b], the mean is (a + b)/2 and the variance is $(b - a)^2/12$.

Here b $-$ a = 6, so the variance = 36/12 = **3** (mean = 5).

Exam tip: The variance depends only on the width of the interval, not on where it lies.

Q5 **Answer: 6.28 (accept 6.22 to 6.34)**

Concept: Angular impulse = change of angular momentum: $T\,t = I(\omega_1 - \omega_2)$.

$\omega_1 = 2\pi \times 300/60 = 31.416$ rad/s and $\omega_2 = 0$.

T = 2 $\times$ 31.416/10 = **6.28 N·m**.

Exam tip: This is the rotational form of F t = m Δv; no need to find the angular deceleration first.

Q6 **Answer: (D) 17/48**

Concept: For a cantilever with a UDL w, measured from the fixed end, $y(x) = w\,x^2(6L^2 - 4Lx + x^2)/(24EI)$.

At x = L/2: $y = w(L^2/4)(6L^2 - 2L^2 + L^2/4)/(24EI) = 17wL^4/(384EI)$.

At x = L: $y = wL^4/(8EI) = 48wL^4/(384EI)$. Ratio = **17/48**, option (D).

Exam tip: 5/16 is the corresponding ratio for a cantilever with a point load at its free end; the answer depends on the type of load.

Q7 **Answer: 1.93 (accept 1.91 to 1.94)**

Concept: The axial force at a distance x from the free end is $\rho g A x$, so $\delta = \int \rho g x\,dx/E = \rho g L^2/(2E)$: half the elongation that the full weight would cause if hung at the end.

$\delta = 7850 \times 9.81 \times 100^2/(2 \times 200 \times 10^9) = 1.925 \times 10^{-3}$ m.

Elongation = **1.93 mm**; it does not depend on the cross-sectional area.

Exam tip: In terms of the total weight W, $\delta = WL/(2AE)$; the maximum stress, $\rho g L$ = 77 MPa here, occurs at the top.

Q8 **Answer: (A) copies a path to an enlarged or reduced scale, its links forming a parallelogram**

Concept: In a pantograph four links form a parallelogram, so a tracing point and a pencil point always move along geometrically similar paths.

It is used to copy drawings and in engraving or copying machines at a fixed scale ratio: option **(A)**.

The other options describe the quick-return mechanism, the Scotch yoke and Oldham's coupling.

Exam tip: Linkages in which a point traces an exact straight line (Peaucellier, Hart) are a separate group from copying linkages such as the pantograph.

Q9 **Answer: (C) 14.6 mm**

Concept: The torque is carried by the shear force in the bolts acting at the pitch-circle radius: $T = n\,F\,(D_p/2)$.

$F = 3000/(6 \times 0.075) = 6667$ N per bolt.

$d = \sqrt{(4F/(\pi\tau))} = \sqrt{(4 \times 6667/(\pi \times 40))} =$ **14.6 mm**, option (C).

Exam tip: 20.6 mm comes from using the pitch-circle diameter instead of the radius; 10.3 mm from dividing the torque by 12.

Q10 **Answer: 4**

Concept: $Re = VD/\nu$; the flow in a pipe stays laminar while $Re \le 2000$.

$V = Re\,\nu/D = 2000 \times 1.0 \times 10^{-4}/0.05 =$ **4 m/s**.

Exam tip: Viscous oils stay laminar at quite high velocities, while water ($\nu \approx 10^{-6}$ m^2/s) in the same pipe turns turbulent above only 0.04 m/s.

Q11 **Answer: (B) 80**

Concept: For an infinitely long fin, $\varepsilon_f = \sqrt{(hPkA)}\theta_b/(hA\theta_b) = \sqrt{(kP/(hA))}$; for a circular pin $P/A = 4/d$.

$\varepsilon_f = \sqrt{(4k/(hd))} = \sqrt{(4 \times 200/(25 \times 0.005))} = \sqrt{6400} =$ **80**, option (B).

Exam tip: Fins pay off when k is high and h is low; they are usually not worth adding if the effectiveness is below about 2.

Q12 **Answer: 347.6 (accept 346.5 to 348.7)**

Concept: With $Q = mc(T_2 - T_1)$ and $\Delta S = mc\,\ln(T_2/T_1)$, $T_m = (T_2 - T_1)/\ln(T_2/T_1)$, the logarithmic mean of the two temperatures.

$T_m = 100/\ln(4/3) = 100/0.2877 =$ **347.6 K**.

Exam tip: T_m is slightly below the arithmetic mean (350 K). Raising the mean temperature of heat addition is why regeneration and superheating improve Rankine-cycle efficiency.

Q13 **Answer: (D) the zone is austenitised and then cooled fast, forming martensite**

Concept: Metal just outside the weld is heated above the upper critical temperature and then quenched by the surrounding cold plate.

In a hardenable steel this fast cooling produces **martensite** (option D), which is hard and brittle and, with hydrogen present, causes cold cracking.

Preheating the plate lowers the cooling rate and avoids this.

Exam tip: The higher the carbon equivalent of the steel, the higher the preheat temperature needed.

Q14 **Answer: 135.7 (accept 134.5 to 136.9)**

Concept: Maximum blanking force $F = \pi\,d\,t\,\tau$; work = F × (penetration distance).

$F = \pi \times 40 \times 3 \times 300 = 113\,097$ N = 113.1 kN.

Penetration = $0.4 \times 3 = 1.2$ mm, so the work = $113\,097 \times 1.2 \times 10^{-3} =$ **135.7 J**.

Exam tip: Providing shear on the punch spreads this same work over a larger stroke, so the peak force falls.

Q15 **Answer: (A) 8.57**

Concept: Multifactor productivity = output ÷ (sum of the input costs considered).

Labour cost = 400 × 200 = ₹80 000; total input = 80 000 + 60 000 = ₹1 40 000.

Productivity = 1200/140 = **8.57 units per ₹1000**, option (A).

Exam tip: Single-factor (partial) productivity uses only one input, e.g. 1200/400 = 3 units per labour-hour.

What next?

Day 87 test tonight, 7 to 11 PM

65 questions · 180 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

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Next, Day 88/90: Mixed PYQ Traps. Comment **GATE90** on our reel to get all links.
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